Machine Learning Assignment 2

Online Shopping Purchase Prediction

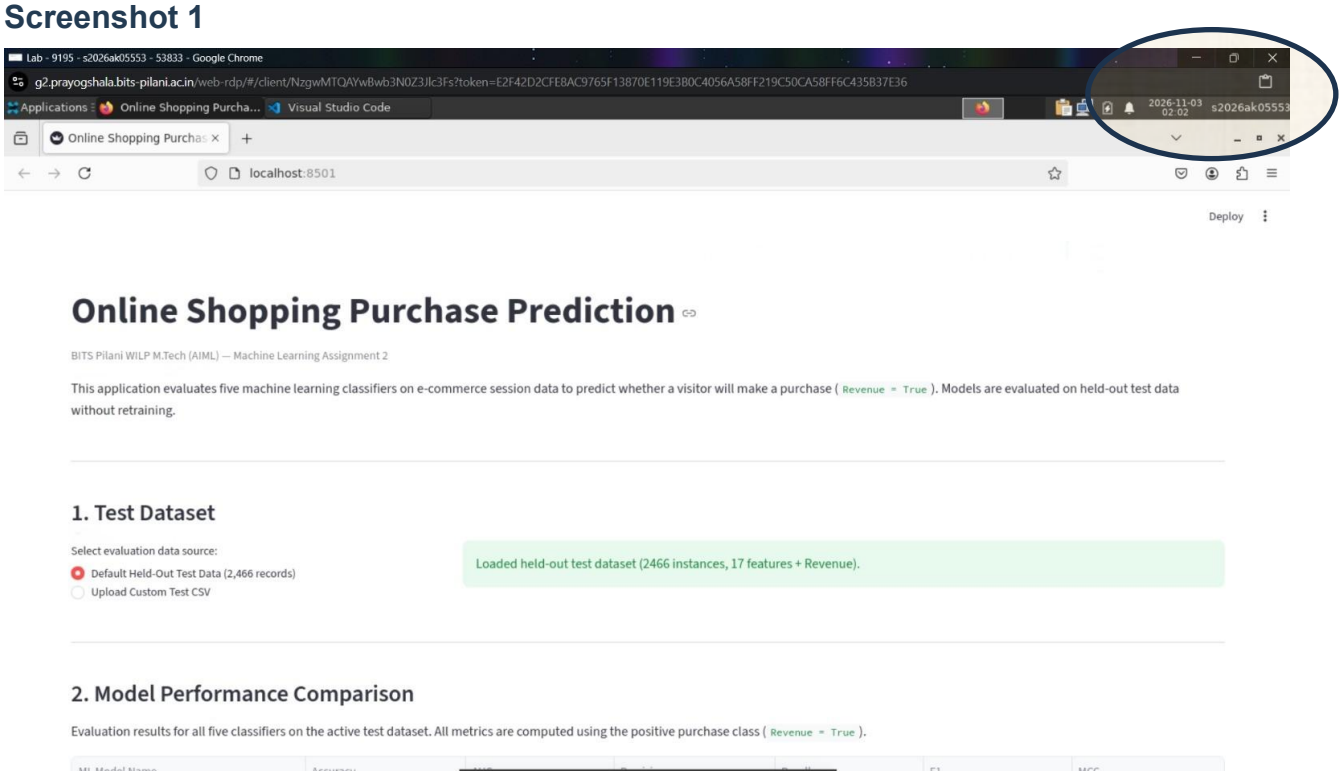
Submission Links

GitHub Repository
https://github.com/Akhil4826/online-shopping-purchase-prediction

Live Streamlit Application
https://online-shopping-purchase-prediction.streamlit.app/

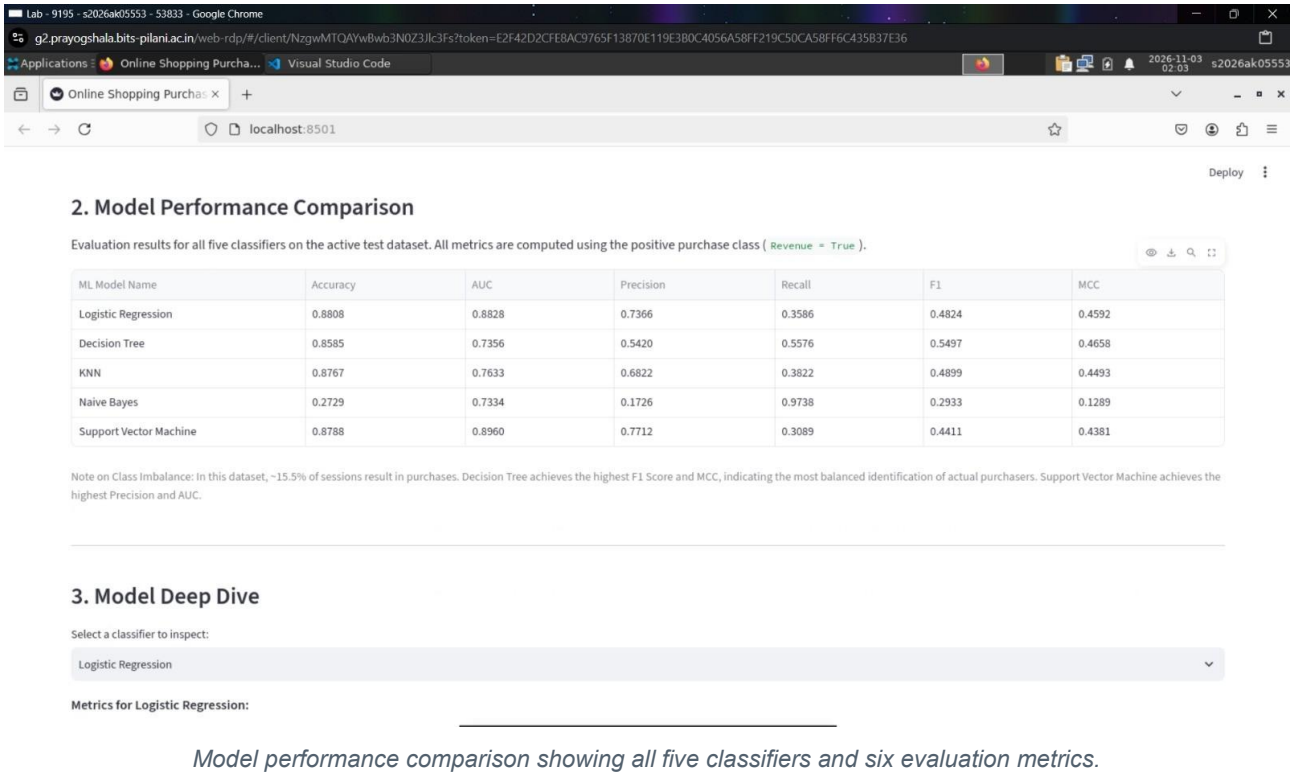
BITS Virtual Lab Execution Screenshots

Screenshot 1



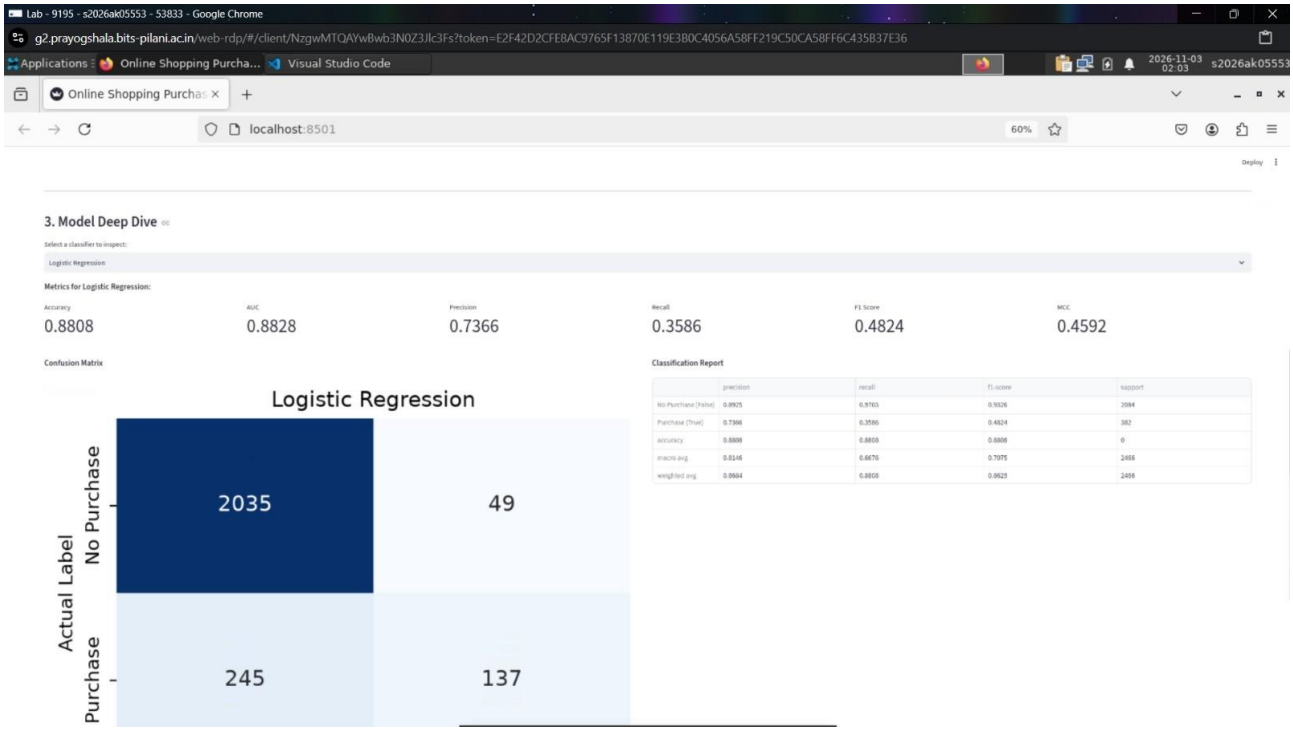
BITS Virtual Lab — Streamlit application opened with the held-out test dataset loaded.

Screenshot 3



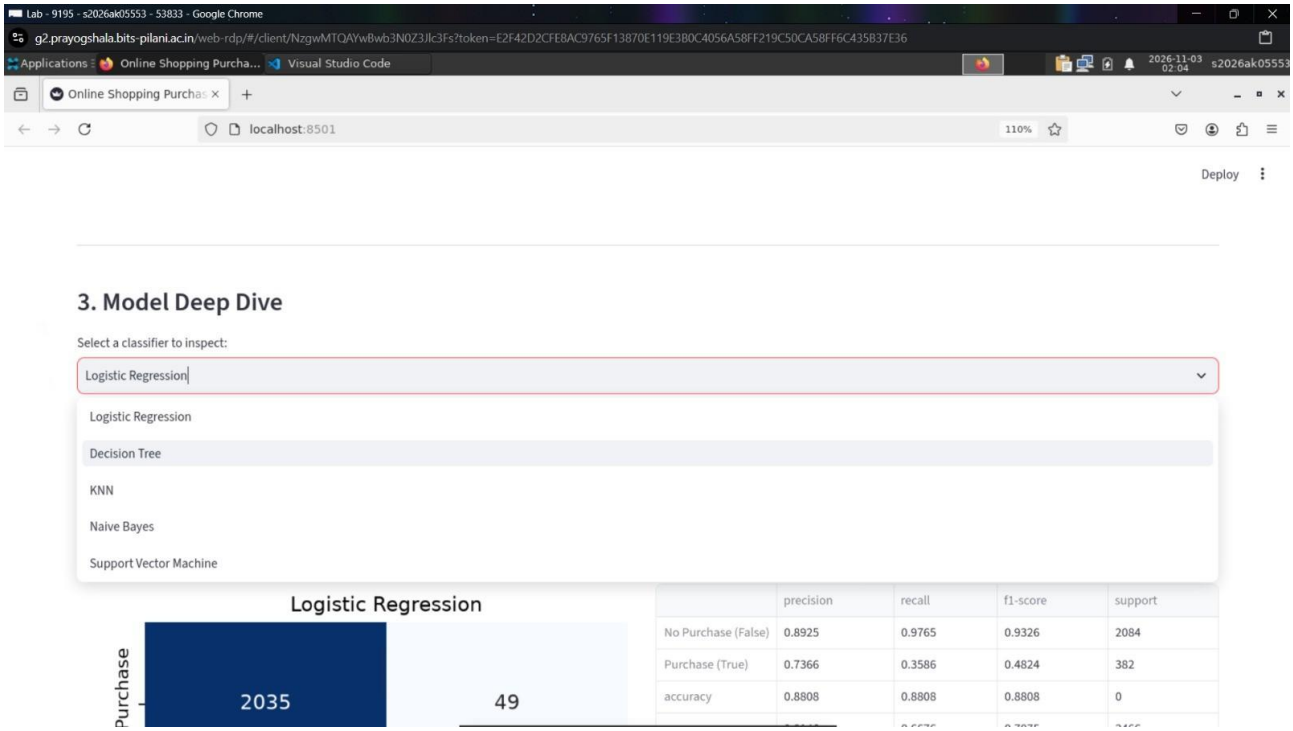
Model performance comparison showing all five classifiers and six evaluation metrics.

Screenshot 3



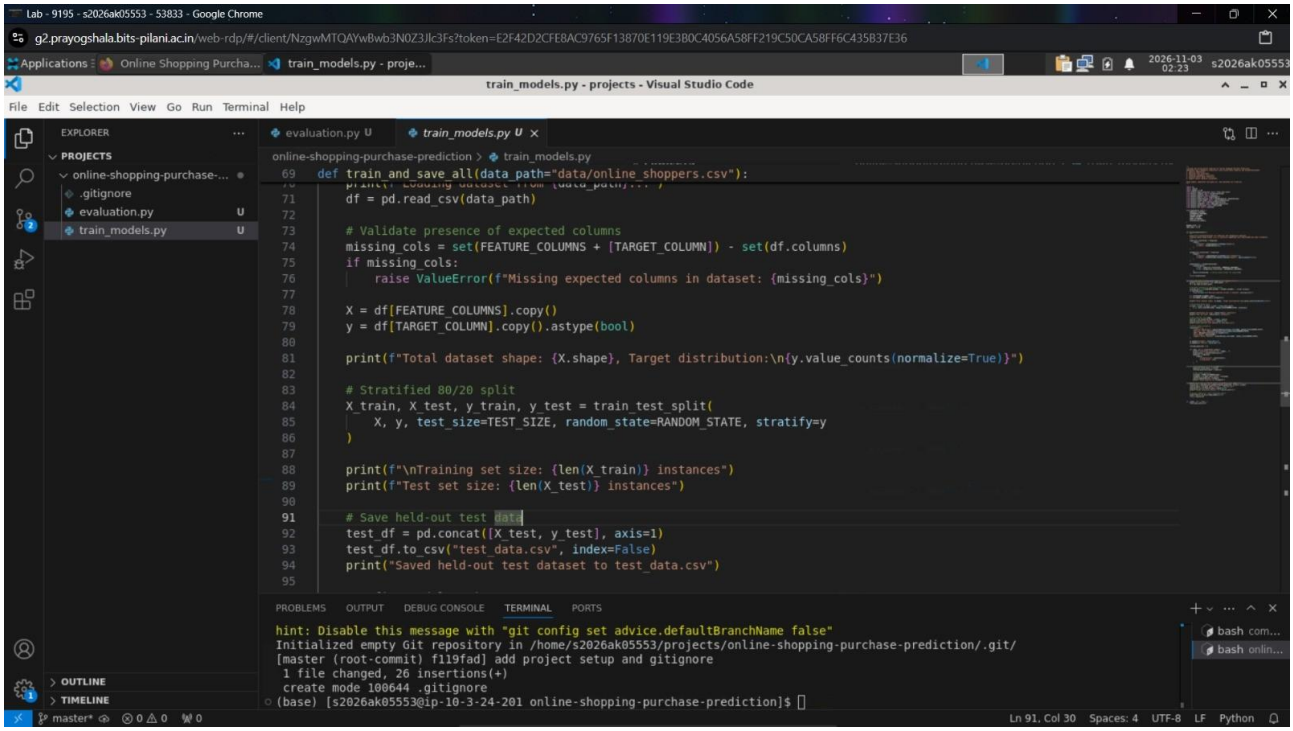
Model Logistic Regression metrics, confusion matrix and classification report.

Screenshot 4



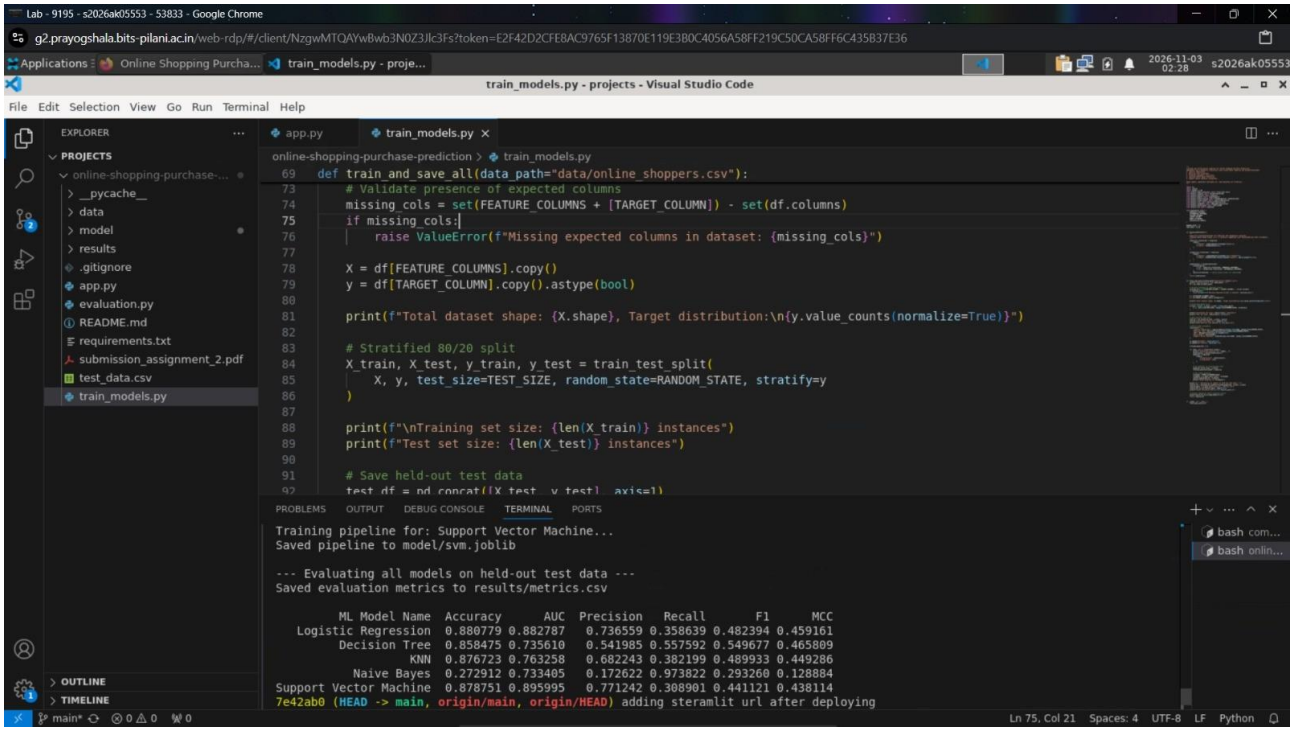
Classifier selection dropdown showing all five available models.

Screenshot 5



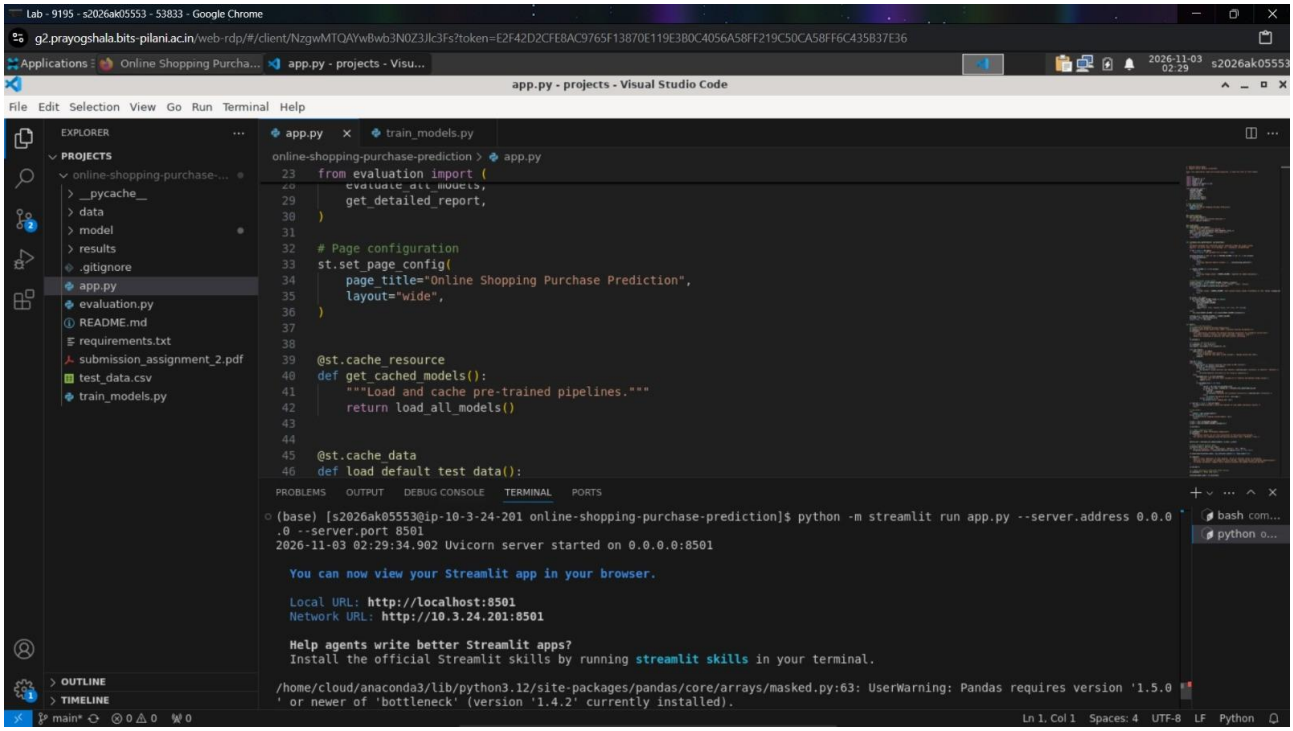
showing the training script and held-out train/test split.

Screenshot 6



Assignment workspace showing model evaluation metrics.

Screenshot 7



Assignment Executed

README

Online Shopping Purchase Prediction

BITS Pilani WILP M.Tech (AIML) — Machine Learning Assignment 2

A. Problem Statement

This project predicts whether a visitor will complete a purchase during an online shopping session using browsing and session behavior data. Five classification models are trained and compared on held-out test data to see which model handles purchase prediction best.

B. Dataset Description

Dataset Name	Online Shoppers Purchasing Intention Dataset
Source	UCI Machine Learning Repository (ID: 468)
Total Records	12,330 sessions
Input Features	17 features (10 numerical features such as durations, bounce rates, exit rates, and page values; 7 categorical features such as month, visitor type, region, and operating system)
Target Variable	Revenue (True = purchase made, False = no purchase)
Class Distribution	10,422 False (84.53%) and 1,908 True (15.47%)

C. GitHub Repository Link

CODE

Live Streamlit Application:

STREAMLIT

D. Models Used

1. **Logistic Regression:** A linear model predicting purchase probability using regularized log-odds.
2. **Decision Tree Classifier:** A tree-based model splitting features recursively by Gini impurity.
3. **K-Nearest Neighbors (KNN):** An instance-based classifier using distances to the 5 nearest neighbors.
4. **Naive Bayes (Gaussian):** A probabilistic classifier applying Bayes' rule assuming feature independence.
5. **Support Vector Machine (LinearSVC):** A linear classifier finding the maximum margin decision boundary.

E. Model Comparison

The table below shows the evaluation results on the 20% held-out test set (2,466 records). All metrics use Revenue = True as the positive class:

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.8808	0.8828	0.7366	0.3586	0.4824	0.4592
Decision Tree	0.8585	0.7356	0.5420	0.5576	0.5497	0.4658
KNN	0.8767	0.7633	0.6822	0.3822	0.4899	0.4493
Naive Bayes	0.2729	0.7334	0.1726	0.9738	0.2933	0.1289
Support Vector Machine	0.8788	0.8960	0.7712	0.3089	0.4411	0.4381

F. Observations

ML Model Name	Observation about model performance
Logistic Regression	Achieved high accuracy (88.08%) and good precision (73.66%), but had low recall (35.86%) because it predicts the positive minority class conservatively.
Decision Tree	Balanced positive-class precision (54.20%) and recall (55.76%), leading to the highest F1 score (0.5497) and highest MCC (0.4658).
KNN	Showed decent overall accuracy (87.67%) and precision (68.22%), but achieved modest recall (38.22%) on the minority class.
Naive Bayes	Detected nearly all buyers (97.38% recall), but produced too many false positives because browsing features are correlated, dropping accuracy to 27.29%.
Support Vector Machine	Achieved the highest AUC (0.8960) and highest precision (77.12%), making it the most reliable model when false positives must be avoided.

G. Overall Winner

Winner: Decision Tree Classifier

Because this dataset is imbalanced (only ~15.5% purchases), overall accuracy is dominated by the negative class. The Decision Tree is the overall winner because it achieves the highest positive-class F1 score (0.5497) and highest MCC (0.4658), identifying over 55% of actual shoppers who buy while maintaining reasonable precision.

Running the Project Locally

```
git clone https://github.com/Akhil4826/online-shopping-purchase-prediction.git
cd online-shopping-purchase-prediction
pip install -r requirements.txt
streamlit run app.py
```